Problem-Solution Fit

Date	26 June 2026
Team ID	
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas Description:

The Problem–Solution Fit Canvas identifies customer needs, challenges, and behaviours while demonstrating how the Smart Lender system effectively addresses them through machine learning-based loan prediction.

It demonstrates how the Smart Lender – Loan Approval Prediction System aligns with the needs of loan applicants and financial institutions. By analysing customer problems, behaviours, limitations, and the root causes of delayed loan assessment, the canvas highlights the importance of an automated prediction system. The proposed solution uses machine learning to provide quick and accurate loan eligibility predictions through a user-friendly web application. This approach reduces manual effort, improves decision-making, and enhances the overall experience for both applicants and financial organizations.

Furthermore, the canvas serves as a strategic planning tool by ensuring that every feature of the Smart Lender system is developed to address real-world user requirements. It also helps identify opportunities for improving system performance, usability, and prediction accuracy based on customer expectations. By establishing a strong connection between the identified problems and the proposed solution, the canvas supports the development of a practical, reliable, and user-centric application that delivers value to both loan applicants and financial institutions.

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Problem-Solution Fit Canvas:-

1. CUSTOMER SEGMENT(S) [CS]

- Loan applicants
- Banks and financial institutions
- Loan officers
- Individuals seeking loan eligibility prediction

6. CUSTOMER LIMITATIONS
EG. BUDGET, DEVICES

- **[CL] Limited knowledge**
- **Internet access**
- **Budget limits**

5. AVAILABLE SOLUTIONS
PROS & CONS **[AS]**
Manual: Human review, slow
Traditional: Organized data, no prediction

2. PROBLEMS / PAINS
+ ITS FREQUENCY **[PR]**

- Manual loan evaluation is slow.
- Loan approval decisions may be inconsistent.
- Applicants are uncertain about loan eligibility.
- High frequency during every loan application.

9. PROBLEM ROOT / CAUSE [RC]

- Manual loan assessment takes time.
- Large number of loan applications.
- Difficulty analyzing multiple applicant factors.
- Lack of instant eligibility prediction

7. BEHAVIOR
+ ITS INTENSITY **[BE]**

- Check eligibility
- Prefer online tools
- Need quick results

3. TRIGGERS TO ACT [TR]

- **Need for quick loan eligibility prediction.**
- **Reduce manual verification time.**
- **Improve decision-making efficiency.**
- **Support applicants before formal loan processing.**

10. YOUR SOLUTION [SL]

- Random Forest prediction
- Flask web application
- Fast, accurate results
- Reduces manual work
- Easy to use
- Instant prediction

8. CHANNELS OF BEHAVIOR [CH]

Online

- Web app
- Bank website

Offline

- Bank branch

4. EMOTIONS
BEFORE / AFTER [
Before: Uncertain, Anxious
After: Confident, Satisfied

Offline

- Bank branch